Examination Answer Evaluation (Case Study 31)

Problem Statement

Compare a student's answers with an answer key and count correct, incorrect, and unanswered questions.

Objectives

Use lists, strings, loops, and counters.

Assumptions

Answers are single letters. A blank answer means unanswered. There is no negative marking in this version.

Algorithm

1. Store the answer key in a list. 2. Read the student's answers. 3. Compare answers position by position. 4. Count correct,
incorrect, and blank answers. 5. Print totals.

Flowchart (Text Form)

START → Set answer key → Read student answers → Compare each answer → Update counters → Print totals →
STOP.

Pseudocode

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```

Python Source Code

```
key = ["A", "B", "C", "D", "A"]
answers = input("Enter 5 answers (use - for blank): ").upper().split()
if len(answers) != 5:
    print("Please enter exactly 5 answers")
else:
    correct = 0
    incorrect = 0
    unanswered = 0
    for i in range(5):
        if answers[i] == "-":
            unanswered += 1
        elif answers[i] == key[i]:
            correct += 1
        else:
            incorrect += 1
    print("Correct:", correct)
    print("Incorrect:", incorrect)
    print("Unanswered:", unanswered)
```

Test Case and Expected Output

Input: A B X - A
Expected output: Correct: 3; Incorrect: 1; Unanswered: 1

Conclusion

The program compares answers with a key and produces a simple result summary.